

Q.17 — Deuteric Alteration of Ca-Rich Plagioclase

DMG 2019 · Igneous Petrology: Secondary/Deuteric Alteration · References: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. & Perkins, *Mineralogy*, 3rd ed.

Q.17. Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?

- (a) Saussuritization
- (b) Devitrification
- (c) Chloritization
- (d) Uralitization

1. The Underlying Concept — What Is Deuteric Alteration?

Winter classifies this whole family of processes under **secondary textures** — changes that occur in an igneous rock after most of it has crystallized, driven by heat and fluid still left over from the magma itself:

"Secondary mineral reactions that occur in igneous rocks as they cool, and that are not products of a distinct later metamorphic event, are commonly called autometamorphic processes rather than metamorphic ones because they are a natural part of igneous cooling."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

"Most, but not all, autometamorphic reactions involve minerals at moderate temperatures in an environment in which H₂O is either liberated from residual melt or externally introduced. Such alterations are a subset of autometamorphism, which involves hydration, and they are called deuteric alterations."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

In short: **deuteric alteration = hydration reactions driven by the magma's own residual water**, acting on minerals that have already crystallized, while the pluton is still cooling. It is a subset of the broader autometamorphic category, and (per Winter) is distinct from diagenesis and weathering, which are explicitly excluded from this category.

2. The Named Deuteric/Secondary Alteration Processes

Winter's Table 1 (Specific Mineral Replacements) gives the definitive one-line summary of each:

Process	Winter's Definition
Saussuritization	"Replacement of plagioclase by epidote."
Uralitization	"Replacement of pyroxene by amphibole."
Biotitization	"Replacement of pyroxene or amphibole by biotite."
Chloritization	"Replacement of any mafic mineral by chlorite."
Sericitization	"Replacement of feldspar or feldspathoids by fine white micas."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 55, Table 1

Devitrification is treated separately, since it doesn't replace one crystalline mineral with another — it involves glass:

"Devitrification is the secondary crystallization of glass to fine-grained mineral aggregates. Glass is an inherently unstable material and is readily replaced by more stable minerals when kinetics permits."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

3. Saussuritization in Full Detail

"Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more

nearly pure albite, releasing Ca and Al to form an epidote mineral ($\pm$ calcite and/or sericite)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 51-52

This is precisely the reaction the question describes: Ca-rich plagioclase $\rightarrow$ nearly pure albite + epidote. Winter even notes where to look for it texturally:

"In zoned plagioclase, one can occasionally observe the products of saussuritization or seritization concentrated in the Ca-rich core or some of the more Ca-rich oscillatory bands in the original grain."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Perkins' *Mineralogy* corroborates this from the mineralogical (product) side, under the epidote entry:

"Saussurite is a name given to fine-grained epidote produced by alteration of plagioclase."

— Perkins, *Mineralogy*, p. 391 (Epidote, Varieties)

"FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar."

— Perkins, *Mineralogy*, p. 391

Ca-rich plagioclase (anorthite component) + $H_2O \rightarrow$ nearly pure albite + epidote/zoisite ($\pm$ calcite, sericite)
The Na-rich (albite) component survives largely unaltered — hence "almost pure albite **and** epidote" as products side by side.

4. Contrast with Uralitization & Chloritization (the Other Deuteric Options)

"Pyroxene is a common primary mafic mineral in a variety of igneous rocks. If H_2O penetrates at modest temperatures, a deuteric alteration of pyroxene to amphibole results, called uralitization. Any gradation from amphibole rims on pyroxene cores to multiple patches of pyroxene in an amphibole to complete replacement is possible."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

"Chloritization is the alteration of any mafic mineral to chlorite. Chlorite is a very hydrous phyllosilicate and typically replaces the less hydrous mafics at low temperature when water is available."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

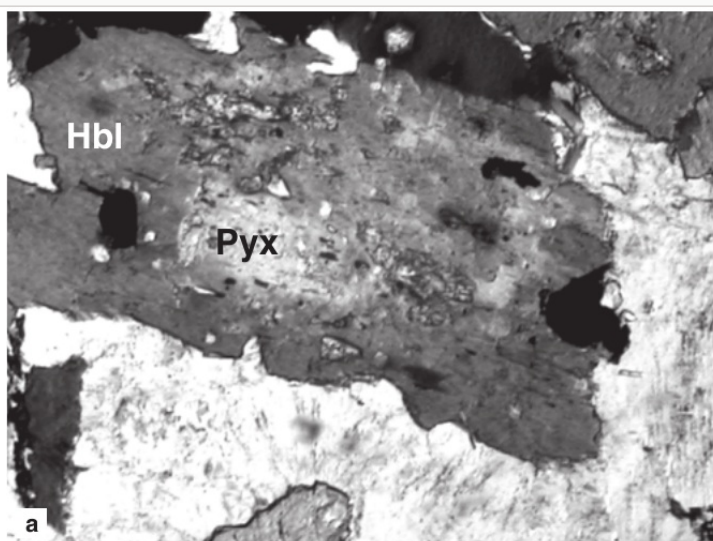




FIGURE 20 (a) Pyroxene largely replaced by hornblende. Some pyroxene remains as light areas (Pyx) in the hornblende core. Width 1 mm. (b) Chlorite (light) replaces biotite (dark) at the rim and along cleavages. Tonalite, San Diego, California. Width 0.3 mm.

generally replaces the initial mafic from the rim inward. In the case of biotite, H_2O may work its way along the prominent cleavages, and chlorite can be seen to replace biotite margins as well as along cleavage planes (Figure 20b).

Sericite is a term applied to any very fine-grained white mica. **Seritization** is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light. In more advanced stages of alteration, the feldspars appear speckled, with fine micas having yellowish birefringence and then larger clots with coarser crystals and higher birefringence. K^+ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K^+ and may be more sericitized than associated plagioclase.

Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite,

Figure 20 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 53) — (a) Pyroxene largely replaced by hornblende (uralitization); remnant pyroxene shown as light areas in the hornblende core. (b) Chlorite (light) replacing biotite (dark) at grain rims and along cleavages (chloritization/biotite-to-chlorite alteration). Neither process involves plagioclase or produces albite + epidote.

octahedra. Small amounts of Ti, Fe, Mg, Na, and K may be present, but no major solid solutions are known.

Occurrence and Associations

Lawsonite is a metamorphic mineral typical of the blueschist facies. Associated high-pressure minerals include glaucophane, jadeite, pumpellyite, or aragonite. Other associated minerals include chlorite, plagioclase, titanite, quartz, and epidote.

Related Minerals

Hemimorphite, $Zn_4(Si_2O_7)(OH)_2 \cdot H_2O$, and ilvaite, $CaFe_3O(Si_2O_7)(OH)$, are other paired tetrahedral silicates in which all SiO_4 tetrahedra are paired. Other minerals usually considered in the same group are epidote, $Ca_2(Al,Fe)_3Si_3O_{12}(OH)$; clinozoisite, $Ca_2Al_3Si_3O_{12}(OH)$; piemontite, $Ca_2(Al,Mn)_3Si_3O_{12}(OH)$; allanite, $(Ca,Ce)_2(Al,Fe,Mg)_3Si_3O_{12}(OH)$; and vesuvianite,



Epidote

$\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$

Origin of Name
From the Greek word *epididonai*, meaning “increase,” referring to the base of an epidote prism, one side of which is longer than the other.

Hand Specimen Identification
Epidote is characterized by its pistachio-green color, habit, one perfect cleavage, and association. It is often fine grained and massive (Figure 19).

Physical Properties

hardness	6 to 7
specific gravity	3.4 to 3.5
cleavage/fracture	perfect (001), poor (100)/uneven
luster/transparency	vitreous/transparent to translucent
color	pistachio-green, yellow-green to black
streak	white



FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar.

The Epidote entry (Perkins, Mineralogy, p. 391), including Figure 19 (a vein of epidote in granite from feldspar alteration) and the Varieties note defining saussurite.

5. Evaluating Each Option

Option	What Actually Alters → Product	Verdict
(a) Saussuritization	Plagioclase → albite + epidote (Winter, p. 55, Table 1) — exact match	Correct
(b) Devitrification	Glass → fine crystalline aggregate (felsite/spherulites); no plagioclase involved	Eliminated
(c) Chloritization	Any mafic mineral → chlorite; not feldspar-related	Eliminated
(d) Uralitization	Pyroxene → amphibole; not feldspar-related	Eliminated

✓ Answer: (a) Saussuritization

6. Probable Follow-Up Exam Questions

Q. What single-line definition would Winter's Table 1 give for saussuritization?

A. "Replacement of plagioclase by epidote."

Q. Why does higher-temperature (Ca-rich) plagioclase break down preferentially over its sodic counterpart?

A. Because Ca-rich plagioclase is less stable than the sodic (albite) component at low temperatures, so on cooling it decomposes, releasing Ca and Al to form epidote while albite persists.

Q. What textural evidence in a zoned plagioclase crystal points to saussuritization?

A. Alteration products concentrated in the Ca-rich core or in Ca-rich oscillatory zoning bands of the original grain (Winter, p. 52).

Q. What is uralite, and how does it form?

A. An amphibole (single hornblende crystal or fibrous actinolite/hornblende aggregate) formed by deuteric alteration of pyroxene; the term is more commonly applied to the fibrous aggregates.

Q. Is saussuritization considered autometamorphic, diagenetic, or metamorphic?

A. Autometamorphic (specifically deuteric) — Winter explicitly excludes diagenesis and weathering from the autometamorphic category, even though the distinction is "rather arbitrary."

Q. What mineral commonly forms alongside epidote when biotite replaces hornblende (a related deuteric process)?

A. Epidote — because biotite contains little Ca, the Ca released during hornblende-to-biotite alteration forms epidote (biotitization, Winter p. 51).